Solutions Manual

Abstract Algebra by Dummit & Foote (3rd ed.)

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Preface

This solutions manual grew out of my effort to study Dummit & Foote thoroughly.

The textbook is famous for placing important results in the exercises rather than in the main body of the text. Because of this, I will summarize any theory-like results at the beginning of each subsection's solutions so they can be referenced easily.

Please reach out to me at `daniel72@postech.ac.kr` if you spot any mistakes or have any suggestions for improvement.

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1 Introduction to Groups

1.1 Basic Axioms and Examples

Skipped problems: $1 \sim 9$, $11 \sim 14$
Problems that need revisiting: 19

10. Let $G = \{g_1, \cdots, g_n\}$ be a finite group and A the group table of G. By definition, $(A)_{ij} = g_i g_j$.

If G were abelian, then for any indices i, j, we have $g_i g_j = g_j g_i$, which implies that $(A)_{ij} = (A)_{ji}$. The choices for i and j were arbitrary, so it follows that A is symmetric.

Conversely, if A were symmetric then for any indices i, j we have $(A)_{ij} = (A)_{ji}$, which in turn implies that $g_i g_j = g_j g_i$, concluding that G is abelian. ■

15. We do this by induction.

Base case: If $n = 1$, then it is evident that $(a_1)^{-1} = a_1^{-1}$.

Inductive step: If $(a_1 a_2 \cdots a_k)^{-1} = a_k^{-1} a_{k-1}^{-1} \cdots a_1^{-1}$ for some integer k, then we see that

$$\begin{aligned}((a_1 a_2 \cdots a_k) a_{k+1})^{-1} &= a_{k+1}^{-1}(a_1 a_2 \cdots a_k)^{-1} \\ &= a_{k+1}^{-1}(a_k^{-1} a_{k-1}^{-1} \cdots a_1^{-1}) \\ &= a_{k+1}^{-1} a_k^{-1} a_{k-1}^{-1} \cdots a_1^{-1},\end{aligned}$$

which completes the proof. ■

16. If $x^2 = 1$, then x has order at most 2 by definition. Conversely, if x has order 1 then $x^2 = 1^2 = 1$, and if x has order 2 then $x^2 = 1$. ■

17. If $|x| = n$, then we have $x^n = 1$. Multiplying by x^{-1} on both sides, we get $x^{n-1} = x^{-1}$. ■

18. For the forward directions, from $xy = yx$ multiply y^{-1} on the left of both sides to deduce $y^{-1}xy = x$, then multiply x^{-1} on the left of both sides to deduce $x^{-1}y^{-1}xy = 1$. For the backward directions, from $x^{-1}y^{-1}xy = 1$ multiply x then y on the left of both sides to retrieve $y^{-1}xy = x$ and then $xy = yx$. ■

19. **a.** This can be done by simple counting.

$$\begin{aligned} x^{a+b} &= \underbrace{x \cdots\cdots\cdots x}_{a+b \text{ times}} \\ &= \underbrace{x \cdots x}_{a \text{ times}} \underbrace{x \cdots x}_{b \text{ times}} \\ &= (\underbrace{x \cdots x}_{a \text{ times}})(\underbrace{x \cdots x}_{b \text{ times}}) \\ &= x^a x^b. \end{aligned}$$

also,

$$\begin{aligned} (x^a)^b &= (\underbrace{x \cdots x}_{a \text{ times}})^b \\ &= \underbrace{(\underbrace{x \cdots x}_{a \text{ times}}) \cdots (\underbrace{x \cdots x}_{a \text{ times}})}_{b \text{ times}} \\ &= \underbrace{x \cdots\cdots x}_{ab \text{ times}} \\ &= x^{ab}. \end{aligned}$$

b. This can be seen by simply observing that $x^{-a}x^a = \underbrace{x^{-1} \cdots x^{-1}}_{a \text{ times}} \underbrace{x \cdots x}_{a \text{ times}} = 1$, and $x^a x^{-a}$ similarly.

c. This seems to be way too much case classification without much reward, so I will skip this at the moment.

20. For any positive integer k, we have $x^k = 1$ if and only if $1 = (x^{-1})^k$, by multiplying appropriate terms in both sides. Now, there are 2 cases: x can have either finite or infinite order.

If x has finite order n, then we have $x^k \neq 1$ for $k = 1, \cdots, n-1$, and this implies that $1 \neq (x^{-1})^k$. Also, we have $x^n = 1$, which implies $1 = (x^{-1})^n$. This shows that x^{-1} has order n.

Now, if x has infinite order, then for any fixed positive integer n we have $x^n \neq 1$. This implies that $1 \neq (x^{-1})^n$, which implies that x^{-1} has infinite order.

This completes the proof. ■

21. If x has odd order in the group, then we have $x^n = 1$ for some odd integer n. Then, $x^{n+1} = x$. Since $n+1$ is even, we can write $n+1 = 2k$ for some integer k. From this we conclude that $x^{2k} = (x^2)^k = x$, completing the proof. ■

22. We first observe that for any positive integer i, we have $(g^{-1}xg)^i = g^{-1}x^i g$. (Technically this must be proven through induction, but we will not do it here.)

If x has finite order n, then we have $x^k \neq 1$ for $k = 1, \cdots, n-1$, and $x^n = 1$. For any $k \in \{1, \cdots, n-1\}$, we have $(g^{-1}xg)^k = g^{-1}x^k g$. If this were 1, then it would imply that $x^k = gg^{-1} = 1$, which must not be true, so we have $g^{-1}x^k g \neq 1$. Also, we observe that $(g^{-1}xg)^n = g^{-1}x^n g = g^{-1}g = 1$. This shows that $g^{-1}xg$ also has order n. If x has infinite order, then we can prove similarly that $g^{-1}xg$ has infinite order too.

To conclude $|ab| = |ba|$ from this, simply let $x = ab$ and $g = a$. ■

23. Observe that $(x^s)^t = x^{st} = x^n$. Observe further two things.

- $(x^s)^k \neq 1$ for $k = 1, \cdots, t-1$, since here $sk < st = n$, and $x^i \neq 1$ for all $1 \leq i < n$.
- $(x^s)^t = 1$, since $(x^s)^t = x^n$.

This concludes that x^s has order t. ■

24. Pick any two elements $a, b \in G$ that commutes. We first handle the obvious case when $n = 0$: $(ab)^0 = a^0 b^0 = 1$.

Following the hint, we will first do induction on positive n. For the base case, if $n = 1$ then $(ab)^1 = a^1 b^1$ is trivially true. For the inductive step, assume that $(ab)^k = a^k b^k$. Then, we have

$$\begin{aligned}
(ab)^{k+1} &= a(ba)^k b \\
&= a(ab)^k b && \text{(since } a \text{ and } b \text{ commute)} \\
&= aa^k b^k b && \text{(by the induction hypothesis)} \\
&= a^{k+1} b^{k+1}.
\end{aligned}$$

This shows that for any positive n, we have $(ab)^n = a^n b^n$.

To show that this fact extends to negative integers, fix an arbitrary positive integer m, and we wish to show that $(ab)^{-m} = a^{-m} b^{-m}$. We first see that since a and b commutes, a^{-1} and b^{-1} also commutes, since $a^{-1}b^{-1} = (ba)^{-1} = (ab)^{-1} = b^{-1}a^{-1}$. Then, we observe that

$$\begin{aligned}
(ab)^{-m} &= ((ab)^{-1})^m \\
&= (b^{-1}a^{-1})^m \\
&= (a^{-1}b^{-1})^m && \text{(since } a^{-1} \text{ and } b^{-1} \text{ commutes)} \\
&= (a^{-1})^m (b^{-1})^m && \text{(again, since } a^{-1} \text{ and } b^{-1} \text{ commutes)} \\
&= a^{-m} b^{-m}.
\end{aligned}$$

This completes the argument. ■

25. $x^2 = 1$ implies that $x = x^{-1}$. Then, for any $a, b \in G$, we see that

$$\begin{aligned}
ab &= a^{-1}b^{-1} && \text{(since } x = x^{-1}\text{)} \\
&= (ba)^{-1} \\
&= ba. && \text{(since } x = x^{-1}\text{)}
\end{aligned}$$

This shows that G is abelian. ■

26. We first see that $\star$ is indeed a binary operation on H, i.e. a function such that $\star : H \times H \to H$, since H is closed under $\star$. (To spell it out, it is because for any $h, k \in H$, we have $h \star k \in H$.)

Now we check the group axioms. Associativity of $\star$ in H is inherited directly from the fact that G is associative. If 1 is the identity of G, then in particular, for any $h \in H$, we have $h \star 1 = 1 \star h = h$. Additionally we have to check that $1 \in H$. To see this, for some $h \in H$ we have $h^{-1} \in H$, so consequently $1 = hh^{-1} \in H$. Finally, since H is closed under inverses, for any $h \in H$ we have $h^{-1} \in H$, so the inverse axiom is satisfied. This completes the proof. ■

27. Denote $\langle x \rangle = \{x^n \mid n \in \mathbb{Z}\}$. It suffices to show that for any two elements $p, q \in \langle x \rangle$, it must be that $pq \in \langle x \rangle$ and $p^{-1} \in \langle x \rangle$.

Noting that $p = x^{n_p}$ for some $n_p \in \mathbb{Z}$ and $q = x^{n_q}$ for some $n_q \in \mathbb{Z}$, it is easy to see that $pq = x^{n_p} x^{n_q} = x^{n_p + n_q} \in \langle x \rangle$ and $p^{-1} = (x^{n_p})^{-1} = x^{-n_p} \in \langle x \rangle$. ■

28. **a.** We have

$$[(a_1, b_1)(a_2, b_2)](a_3, b_3) = ([a_1 a_2]a_3, [b_1 b_2]b_3)$$

and

$$(a_1, b_1)[(a_2, b_2)(a_3, b_3)] = (a_1[a_2 a_3], b_1[b_2 b_3]).$$

The associativity of groups A and B imply that the two expressions the same, so that $A \times B$ is associative.

b. For any $a \in A$ and $b \in B$, we have $(1, 1)(a, b) = (1a, 1b) = (a, b)$ and $(a, b)(1, 1) = (a1, b1) = (a, b)$, implying that $(1, 1)$ is the identity of $A \times B$.

c. For any $a \in A$ and $b \in B$, if a^{-1} is the inverse of a in A and b^{-1} the inverse of b in B, then we see that $(a, b)(a^{-1}, b^{-1}) = (aa^{-1}, bb^{-1}) = (1, 1)$ and $(a^{-1}, b^{-1})(a, b) = (a^{-1}a, b^{-1}b) = (1, 1)$, showing that (a^{-1}, b^{-1}) is the inverse of (a, b) in $A \times B$. ■

29. $A \times B$ is an abelian group if and only if for any elements $a_1, a_2 \in A$ and $b_1, b_2 \in B$, we have $(a_1, b_1)(a_2, b_2) = (a_2, b_2)(a_1, b_1)$. Here, $(a_1, b_1)(a_2, b_2) = (a_1 a_2, b_1 b_2)$ and $(a_2, b_2)(a_1, b_1) = (a_2 a_1, b_2 b_1)$. This is true if and only if $a_1 a_2 = a_2 a_1$ and $b_1 b_2 = b_2 b_1$, if and only if A and B are both abelian. ■

30. We see that $(a, 1)(1, b)$ and $(1, b)(a, 1)$ both evaluate to (a, b), so they commute. To see that the order of (a, b) is the lcm of $|a|$ and $|b|$, we observe that

$$\begin{aligned}
(a, b)^n &= [(a, 1)(1, b)]^n \\
&= (a, 1)^n (1, b)^n \\
&= (a^n, 1)(1, b^n) \\
&= (a^n, b^n),
\end{aligned}$$

since $(a, 1)$ and $(1, b)$ commute. For this to be the identity we see that both a^n and b^n must be the identity. a^n is the identity if and only if $|a|$ divides n, and b^n is the identity if and only if $|b|$ divides n. Now, the smallest positive n such that both $|a|$ and $|b|$ divides n, must be the lcm of $|a|$ and $|b|$, and therefore $|(a, b)|$ must be the lcm of $|a|$ and $|b|$. ■

31. Let G be a group of even order. We have to prove that there is an element in G of order 2.

Following the hint, define the set $t(G) = \{g \in G \mid g \neq g^{-1}\}$. We first see that if some a is some element in G such that $a \in t(G)$, then it must be that $a^{-1} \in t(G)$ since $(a^{-1})^{-1} \neq a^{-1}$. Moreover, a and a^{-1} must be distinct, again because of the condition $a \neq a^{-1}$. From this we can argue that $t(G)$ as a set has even order, since the elements in $t(G)$ can be paired in two: (a, a^{-1}). Since G also has even order, we conclude that $G \setminus t(G)$ must also have even order.

Now, we see that $x \in G \setminus t(G)$ if and only if $x = x^{-1}$. This implies that $x^2 = 1$, so if x is not the identity then x has order 2 in G. Also, since $1 = 1^{-1}$, we have $1 \in G \setminus t(G)$. Since $G \setminus t(G)$ has even order, some other element in $G \setminus t(G)$, other than the identity, must exist, and this element must have order 2 in G. This concludes the proof. ■

32. By definition of an order of an element, if n is the order of the element x, we have $x^k \neq 1$ for $k = 1, \cdots, n-1$, and $x^n = 1$. Now, assume for the sake of contradiction that $x^i = x^j$ for some $1 \leq i < j \leq n-1$. Then, $x^i x^{n-j} = x^j x^{n-j}$ implies $x^{n+i-j} = x^n$. Observing that $0 < n+i-j < n$ and $x^n = 1$, we see that $x^{n+i-j} = 1$, which contradicts our very first observation. Therefore, $1, x, \cdots, x^{n-1}$ must be distinct elements.

To deduce $|x| \leq |G|$ from here, we simply observe that G has at least n elements from the previous paragraph. ■

33. Assume x is of order n in the group G. We have $x^n = 1$ and $x^k \neq 1$ for $k = 1, \cdots, n-1$. Then, for $1 \leq j < 2n$, the only value of j that satisfies $x^j = 1$ is $j = n$.

a. If n is odd, then for $1 \leq j < 2n$ the only value that satisfies $x^j = 1$ is n, which is odd.

Now, suppose for the sake of contradiction that for some i such that $1 \leq i < n$, we have $x^i = x^{-i}$. Then, we have $x^{2i} = 1$. But $1 \leq 2i < 2n$ and $2i$ is even, contradicting our initial observation. Therefore, if n is odd then it must be that $x^i \neq x^{-i}$ for $i = 1, \cdots, n-1$.

b. If n is even, then for $1 \leq j < 2n$ the only value that satisfies $x^j = 1$ is n, and if this is the case then $j = n = 2k$.

Here, suppose that for some i such that $1 \leq i < n$, we have $x^i = x^{-i}$. Then, we have $x^{2i} = 1$. But since $1 \leq 2i < 2n$, it must be the case that $2i = 2k$, or that $i = k$. Conversely, we obviously have $x^k = x^{-k}$ since $x^{2k} = x^n = 1$.

This completes the argument. ■

34. x being an infinite order element in G implies that for any positive integer n, we have $x^n \neq 1$. By similar logic from the solution for problem 32, for the sake of contradiction assume that $x^i = x^j$ for some distinct integers i and j, and assume $i < j$ without loss of generality. Then, $x^i x^{-i} = x^j x^{-i}$ implies that $x^{j-i} = 1$, which contradicts our very first observation. Therefore, elements of the form x^n where $n \in \mathbb{Z}$ must be distinct. ■

35. By the definition of the order of an element in a group, if x has finite order n in G then we have distinct elements $1, x, \ldots, x^{n-1} \in G$, and then $x^n = 1$.

Now, fix any $a \in \mathbb{Z}$, and we are left to show that $x^a \in \{1, x, \cdots, x^{n-1}\}$. By the division algorithm, we can write $a = qn + r$ for some unique q and r where $0 \leq r < n$. Then we observe that

$$\begin{aligned} x^a &= x^{qn+r} \\ &= (x^n)^q x^r \\ &= x^r \in \{1, x, \cdots, x^{n-1}\}, \end{aligned}$$

since $0 \leq r < n$. This completes our argument. ■

36. The contraposition of the cancellation laws state that for $u, v \in G$, if $u \neq v$ then we have $au \neq av$ and $ua \neq va$ for any $a \in G$. This implies that the group table must not have duplicate elements in the same row or column. Keeping this rule in mind, and ensuring that there are no elements of order 4, we see that the table below is the only possible group table for $G = \{1, a, b, c\}$.

$\cdot$	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

This matrix is symmetric, so by Exercise 1.1.10 G must be abelian. ■

(Note: we haven't proved that the group axioms for the binary operation hold for that "group table". We can see that the identity and inverse axioms hold, but associativity is trickier to see. However, the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ is isomorphic to this table, and we can deduce associativity from there.)

1.2 Dihedral Groups

Skipped problems: 17, 18
Recall that the elements of D_{2n} are precisely $1, r, r^2, \cdots, r^{n-1}, s, sr, sr^2, \cdots, sr^{n-1}$.

1. **a.** We calculate each element of D_6.

- 1 has order 1.
- r has order 3, illustrated in $r \to r^2 \to 1_{=r^3}$.
- r^2 has order 3, illustrated in $r^2 \to r_{=r^4} \to 1_{=r^3}$.
- s has order 2, illustrated in $s \to 1_{=s^2}$.
- sr has order 2, illustrated in $sr \to 1_{=srsr=s^2}$.
- sr^2 has order 2, illustrated in $sr^2 \to 1_{=sr^2sr^2=s^2}$.

b. We calculate each element of D_8.

- 1 has order 1.
- r has order 4, illustrated in $r \to r^2 \to r^3 \to 1_{=r^4}$.
- r^2 has order 2, illustrated in $r^2 \to 1_{=r^4}$.
- r^3 has order 4, illustrated in $r^3 \to r^2_{=r^6} \to r_{=r^5} \to 1_{=r^4}$.
- s has order 2, illustrated in $s \to 1_{=s^2}$.
- sr has order 2, illustrated in $sr \to 1_{=srsr=s^2}$.
- sr^2 has order 2, illustrated in $sr^2 \to 1_{=sr^2sr^2=s^2}$.
- sr^3 has order 2, illustrated in $sr^3 \to 1_{=sr^3sr^3=s^2}$.

c. We calculate each element of D_{10}.

- 1 has order 1.
- r has order 5, illustrated in $r \to r^2 \to r^3 \to r^4 \to 1_{=r^5}$.
- r^2 has order 5, illustrated in $r^2 \to r^4 \to r_{=r^6} \to r^3 \to 1_{=r^5}$.
- r^3 has order 5, illustrated in $r^3 \to r_{=r^6} \to r^4 \to r^2_{=r^7} \to 1_{=r^5}$.
- r^4 has order 5, illustrated in $r^4 \to r^3_{=r^8} \to r^2_{=r^7} \to r_{=r^6} \to 1_{=r^5}$.
- s has order 2, illustrated in $s \to 1_{=s^2}$.
- sr has order 2, illustrated in $sr \to 1_{=srsr=s^2}$.
- sr^2 has order 2, illustrated in $sr^2 \to 1_{=sr^2sr^2=s^2}$.
- sr^3 has order 2, illustrated in $sr^3 \to 1_{=sr^3sr^3=s^2}$.
- sr^4 has order 2, illustrated in $sr^4 \to 1_{=sr^4sr^4=s^2}$.

2. If x is an element of D_{2n} not a power of r, then it must be of the form sr^i where $0 \leq i < n$. Then, $rx = rsr^i = sr^{-1}r^i = sr^ir^{-1} = xr^{-1}$. ■

3. If x is an element of D_{2n} not a power of r, then it must be of the form sr^i where $0 \leq i < n$. Then, since $r^is = s(r^{-1})^i$, we have $x^2 = sr^isr^i = s^2 = 1$. ($r^is = s(r^{-1})^i$ technically has to be proven inductively, but we will not do this here.) Since $x^1 \neq 1$, we conclude that x has order 2.

Now, any element of D_{2n} can be written as a product of s and sr, since $r = s \cdot sr$ and every element can be written as a product of s and r. Therefore $\{s, sr\}$ is a generator of D_{2n}. s has order 2 as we already know, and sr has order 2 by our first part. ■

4. z is an element of order 2 since $z \neq 1$ and $z^2 = r^{2k} = r^n = 1$. To see that z commutes with every element of D_{2n}, fix some $d \in D_{2n}$, and our goal is to show that $zd = dz$. Consider the two cases:

i. d is a power of r.
Then, $d = r^l$ for some $l \in \mathbb{Z}$, and so

$$\begin{aligned} zd = r^k r^l &= r^{k+l} \\ &= r^{l+k} \qquad \text{(because of commutativity in } \mathbb{Z}) \\ &= r^l r^k = dz, \end{aligned}$$

and therefore $zd = dz$.

ii. d is not a power of r.
Then, by Exercise 1.2.2, we have $rd = dr^{-1}$. Then, we see that

$$\begin{aligned} zd &= r^k d \\ &= r^{k-1} dr^{-1} \\ &= r^{k-2} dr^{-2} \\ &\vdots \\ &= dr^{-k} \\ &= dz^{-1}. \end{aligned}$$

Here, since z is an element of order 2 we have $z^2 = 1$, so that $z = z^{-1}$. Therefore, we can conclude that $zd = dz$.

For every case of d, we have $zd = dz$, and therefore z commutes with every element of D_{2n}.

Finally, we must show that z is the *only* nonidentity that commutes with every element of D_{2n}. To do this, suppose x is some element of D_{2n} that commutes with every element. We will show that then x must be either 1 or r^k. There are two possible cases for the form of x.

i. x is a power of r, so $x = r^i$ where $i \in \{0, 1, \cdots, n-1\}$.
From the logic of part ii above, if d is not a power of r then we must have $xd = dx^{-1}$. However, since x commutes with every element of D_{2n}, it also commuts with d in particular so that $xd = dx$. Therefore we have $dx^{-1} = dx$ and so $x^{-1} = x$.
By Exercise 1.1.33, since r is of even order ($= 2k$) in D_{2n}, it must be that $x^{-1} = x$ if and only if $x = 1$ or $x = r^k$. Therefore, if x is a power of r, then x commutes with every element of D_{2n} if and only if $x = 1$ or $x = r^k$.

ii. x is not a power of r, so $x = sr^i$ where $i \in \{0, 1, \cdots, n-1\}$.
Then, we must have $xr = rx$ in particular. Here, we observe that $xr = sr^i r = sr^{i+1}$ and $rx = rsr^i = sr^{-1}r^i = sr^{i-1}$, and therefore $sr^{i+1} = sr^{i-1}$. So, we must have $r^{i+1} = r^{i-1}$. But this implies that $r^2 = 1$, which says that r has order at most 2. This contradicts the premise that $n \geq 4$. Therefore, in this case there is no element that commutes with every element in D_{2n}.

This concludes the proof. ■

5. The solution is very similar to the latter half of the solution of Exercise 1.2.4.

We must show that if x is some element of D_{2n} that commutes with every element, then it must be that $x = 1$. There are two possible cases for the form of x.

i. x is a power of r, so $x = r^i$ where $i \in \{0, 1, \cdots, n-1\}$.
From the logic of the first part ii in Exercise 1.2.4, if d is not a power of r then we must have $xd = dx^{-1}$. However, since x commutes with every element of D_{2n}, it also commutes with d in particular so that $xd = dx$. Therefore we have $dx^{-1} = dx$ and so $x^{-1} = x$.
By Exercise 1.1.33, since r is of odd order in D_{2n}, it must be that $x^{-1} = x$ if and only if $x = 1$. Therefore, if x is a power of r, then x commutes with every element of D_{2n} if and only if $x = 1$.

ii. x is not a power of r, so $x = sr^i$ where $i \in \{0, 1, \cdots, n-1\}$.
Then, we must have $xr = rx$ in particular. Here, we observe that $xr = sr^ir = sr^{i+1}$ and $rx = rsr^i = sr^{-1}r^i = sr^{i-1}$, and therefore $sr^{i+1} = sr^{i-1}$. So, we must have $r^{i+1} = r^{i-1}$. But this implies that $r^2 = 1$, which says that r has order at most 2. This contradicts the premise that $n \geq 3$. Therefore, in this case there is no element that commutes with every element in D_{2n}.

This concludes the proof. ■

6. Let x and y be elements of order 2 in G. We essentially only need to prove that $xyx = xy^{-1}x^{-1}$, which is definitely true since $x^2 = 1$ and $y^2 = 1$ so that $x = x^{-1}$ and $y = y^{-1}$. ■

(The book seems to want to say that there is a parallel between $x \leftrightarrow s$ and $y \leftrightarrow sr$, similar to what we've seen in Exercise 1.2.3. Then we have $t = xy \leftrightarrow ssr = r$, so x and t satisfy the same relations as s and r do in D_{2n}.)

7. Recall that $D_{2n} = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$ and $C_{2n} = \langle a, b \mid a^2 = b^2 = (ab)^n = 1 \rangle$. We show that these are two presentations of the same group by changing generators.

First start with the presentation of D_{2n}, and define new generators $a := s$ and $b := sr$. Then $s = a$, and also $r = ssr = ab$. Now compute:

- $a^2 = s^2 = 1$,
- $b^2 = (sr)^2 = srsr = ssr^{-1}r = 1$,
- $(ab)^n = (s \cdot sr)^n = r^n = 1$.

Thus, in terms of the generators a, b, the defining relations for D_{2n} become $a^2 = b^2 = (ab)^n = 1$.

Conversely, start with the presentation of C_{2n}, and define $s := a$ and $r := ab$. Then $a = s$, and $b = a(ab) = sr$. Now compute:

- $r^n = (ab)^n = 1$,
- $s^2 = a^2 = 1$,
- $rs = (ab)a = aba = a(ba) = a(ab)^{-1} = sr^{-1}$, since $a^2 = b^2 = 1$, so $ba = (ab)^{-1}$.

Thus, in terms of the generators r, s, the defining relations for C_{2n} become $r^n = s^2 = 1$ and $rs = sr^{-1}$.

Therefore each presentation is obtained from the other by a change of generators, so C_{2n} and D_{2n} are two presentations of the same group. ■

8. We determine the order of the subgroup $\langle r \rangle = \{r^k \mid k \in \mathbb{Z}\}$. Since $|r| = n$ in D_{2n}, the elements $1, r, r^2, \cdots, r^{n-1}$ are all distinct, and $r^n = 1$. Hence every power of r is equal to one of $1, r, r^2, \cdots, r^{n-1}$, and these are exactly the elements of $\langle r \rangle$. Therefore, $|\langle r \rangle| = n$. ■

For exercises 9 through 13, we follow the logic spelled out below.

For each Platonic solid, fix an ordered pair (u_1, u_2) of adjacent vertices. A rigid motion is completely determined by the image of (u_1, u_2). Indeed, once the image of one vertex and of an adjacent vertex is fixed, the rest of the solid is forced.

Now suppose the solid has v vertices, and each vertex is adjacent to e vertices. The vertex u_1 can be sent to any of the v vertices. After that, u_2 can be sent to any of the e vertices adjacent to the image of u_1. Since the solid is regular, every such ordered adjacent pair is realized by a rigid motion. Therefore, the number of rigid motions is ve.

9. A tetrahedron has 4 vertices and 3 edges per vertex. Therefore, $|G| = 4 \times 3 = 12$. ■

10. A cube has 8 vertices and 3 edges per vertex. Therefore, $|G| = 8 \times 3 = 24$. ■

11. An octahedron has 6 vertices and 4 edges per vertex. Therefore, $|G| = 6 \times 4 = 24$. ■

12. A dodecahedron has 20 vertices and 3 edges per vertex. Therefore, $|G| = 20 \times 3 = 60$. ■

13. An icosahedron has 12 vertices and 5 edges per vertex. Therefore, $|G| = 12 \times 5 = 60$. ■

14. As mentioned in the text, $\mathbb{Z} = \langle 1 \rangle$ works, since any integer can be written as a sum of a finite number of 1's or -1's. ■

15. $\mathbb{Z}/n\mathbb{Z} = \langle 1 \mid n \cdot 1 = 0 \rangle$ is a possible presentation.

To justify that this presentation works, observe first that every element of $\mathbb{Z}/n\mathbb{Z}$ can be written as $k \cdot 1$ for some $k \in \mathbb{Z}$, so 1 generates the group. Also, in $\mathbb{Z}/n\mathbb{Z}$ we have $n \cdot 1 = 0$, since $n \equiv 0 \pmod{n}$.

Finally, this is the only relation needed: whenever $a \cdot 1 = b \cdot 1$, we have $(a - b) \cdot 1 = 0$, which in $\mathbb{Z}/n\mathbb{Z}$ means exactly that $a - b$ is divisible by n. Therefore two integer multiples of 1 are equal precisely when they become equal using the relation $n \cdot 1 = 0$. Hence $\langle 1 \mid n \cdot 1 = 0 \rangle$ is indeed a presentation of $\mathbb{Z}/n\mathbb{Z}$. ■

16. This follows immediately from Exercise 1.2.7.

1.3 Symmetric Groups

Skipped problems: 12∼14, 16∼20

1.

$$\begin{aligned} \sigma &= (1\ 3\ 5)\,(2\ 4), & \tau &= (1\ 5)\,(2\ 3), \\ \sigma^2 &= (1\ 5\ 3), & \sigma\tau &= (2\ 5\ 3\ 4), \\ \tau\sigma &= (1\ 2\ 4\ 3), & \tau^2\sigma &= (1\ 3\ 5)\,(2\ 4). \end{aligned}$$

2.

$$\begin{aligned} \sigma &= (1\ 13\ 5\ 10)\,(3\ 15\ 8)\,(4\ 14\ 11\ 7\ 12\ 9), \\ \tau &= (1\ 14)\,(2\ 9\ 15\ 13\ 4)\,(3\ 10)\,(5\ 12\ 7)\,(8\ 11), \\ \sigma^2 &= (1\ 5)\,(3\ 8\ 15)\,(4\ 11\ 12)\,(7\ 9\ 14)\,(10\ 13), \\ \sigma\tau &= (1\ 11\ 3)\,(2\ 4)\,(5\ 9\ 8\ 7\ 10\ 15)\,(13\ 14), \\ \tau\sigma &= (1\ 4)\,(2\ 9)\,(3\ 13\ 12\ 15\ 11\ 5)\,(8\ 10\ 14), \\ \tau^2\sigma &= (1\ 2\ 15\ 8\ 3\ 4\ 14\ 11\ 12\ 13\ 7\ 5\ 10). \end{aligned}$$

3. As suggested in the book (and proved in Exercise 1.3.15), the order of a permutation is the least common multiple of the lengths of the cycles in its cycle decomposition.

	σ	τ	σ^2	$\sigma\tau$	$\tau\sigma$	$\tau^2\sigma$
1.	6	2	3	4	4	6
2.	12	30	6	6	6	13

4. We use two facts.

First, every permutation can be written uniquely as a product of disjoint cycles, up to reordering the cycles and cyclically permuting the entries within each cycle.

Second, the order of a permutation is the least common multiple of the lengths of the disjoint cycles in its cycle decomposition (by Exercise 1.3.15).

So the strategy is to list all possible disjoint-cycle decompositions in S_3 and S_4, and then compute the order from the cycle lengths.

a. In S_3, every element is one of the following:

- the identity id, which has order 1;
- a transposition: $(1\ 2)$, $(1\ 3)$, $(2\ 3)$, each of which has order 2;
- a 3-cycle: $(1\ 2\ 3)$, $(1\ 3\ 2)$, each of which has order 3.

b. In S_4, every element is one of the following forms:

- the identity id, which has order 1;
- a transposition:
$$(1\ 2),\ (1\ 3),\ (1\ 4),\ (2\ 3),\ (2\ 4),\ (3\ 4),$$
each of which has order 2;
- a 3-cycle:
$$(1\ 2\ 3),\ (1\ 3\ 2),\ (1\ 2\ 4),\ (1\ 4\ 2),$$
$$(1\ 3\ 4),\ (1\ 4\ 3),\ (2\ 3\ 4),\ (2\ 4\ 3),$$
each of which has order 3;
- a product of two disjoint transpositions:
$$(1\ 2)(3\ 4),\ (1\ 3)(2\ 4),\ (1\ 4)(2\ 3),$$
each of which has order 2;
- a 4-cycle:
$$(1\ 2\ 3\ 4),\ (1\ 2\ 4\ 3),\ (1\ 3\ 2\ 4),$$
$$(1\ 4\ 2\ 3),\ (1\ 3\ 4\ 2),\ (1\ 4\ 3\ 2),$$
each of which has order 4.

For a sanity check, we see that these are indeed all of the elements of S_4, since the total number listed is
$$1 + 6 + 8 + 3 + 6 = 24 = 4! = |S_4|.$$

5. Since the cycles are pairwise disjoint, this is a cycle decomposition of some permutation, and therefore the order of this permutation must be the lcm of the lengths of the cycles. (Exercise 1.3.15) So, the order of this permutation is $\operatorname{lcm}(5, 2, 3, 2) = 30$. ■

6. This is done in Exercise 1.3.4(b).

7. This is done in Exercise 1.3.4(b).

8. For a positive integer n, let σ_n be the permutation in S_Ω such that $\sigma_n(n) = 1, \sigma_n(1) = n$, and $\sigma_n(i) = i$ for all other i. Then we see that if $m \neq n$, then $\sigma_n \neq \sigma_m$, which shows that S_Ω has at least $|\mathbb{Z}^+|$-many elements. ■

9. We determine in each case, for which integers i the power is again a single cycle of the same length. Since $\sigma_0^t = \mathrm{id}$ for a t-cycle σ_0, it suffices to consider exponents modulo t. Also, $\sigma_0^{t-i} = (\sigma_0^i)^{-1}$, so σ_0^{t-i} has the same cycle lengths as σ_0^i. Thus it is enough to compute only the first half of the powers.

a. Let

$$\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12).$$

Its first few powers are

$$\begin{aligned}
\sigma^2 &= (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12),\\
\sigma^3 &= (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12),\\
\sigma^4 &= (1\ 5\ 9)(2\ 6\ 10)(3\ 7\ 11)(4\ 8\ 12),\\
\sigma^5 &= (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8),\\
\sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12).
\end{aligned}$$

From these, and using $\sigma^{12-i} = (\sigma^i)^{-1}$, we see that σ^i is a 12-cycle precisely for

$$i \equiv 1, 5, 7, 11 \pmod{12}.$$

b. Let

$$\tau = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8).$$

Then

$$\begin{aligned}
\tau^2 &= (1\ 3\ 5\ 7)(2\ 4\ 6\ 8),\\
\tau^3 &= (1\ 4\ 7\ 2\ 5\ 8\ 3\ 6),\\
\tau^4 &= (1\ 5)(2\ 6)(3\ 7)(4\ 8).
\end{aligned}$$

Hence τ^i is an 8-cycle precisely for

$$i \equiv 1, 3, 5, 7 \pmod{8}.$$

c. Let

$$\omega = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12\ 13\ 14).$$

Its first several powers are

$$\begin{aligned}
\omega^2 &= (1\ 3\ 5\ 7\ 9\ 11\ 13)(2\ 4\ 6\ 8\ 10\ 12\ 14),\\
\omega^3 &= (1\ 4\ 7\ 10\ 13\ 2\ 5\ 8\ 11\ 14\ 3\ 6\ 9\ 12),\\
\omega^4 &= (1\ 5\ 9\ 13\ 3\ 7\ 11)(2\ 6\ 10\ 14\ 4\ 8\ 12),\\
\omega^5 &= (1\ 6\ 11\ 2\ 7\ 12\ 3\ 8\ 13\ 4\ 9\ 14\ 5\ 10),\\
\omega^6 &= (1\ 7\ 13\ 5\ 11\ 3\ 9)(2\ 8\ 14\ 6\ 12\ 4\ 10),\\
\omega^7 &= (1\ 8)(2\ 9)(3\ 10)(4\ 11)(5\ 12)(6\ 13)(7\ 14).
\end{aligned}$$

Therefore ω^i is a 14-cycle precisely for

$$i \equiv 1, 3, 5, 9, 11, 13 \pmod{14}.$$

We can see a pattern: if σ_0 is a t-cycle, then σ_0^i is again a t-cycle exactly when i is relatively prime to t. This will be proved in Exercise 1.3.11.

10. If $\sigma = (a_1\ a_2\ \cdots\ a_m)$, then we have

$$\begin{aligned} \sigma(a_k) &= a_{k+1} \quad \text{for } k = 1, \ldots, m-1, \text{ and} \\ \sigma(a_m) &= a_1. \end{aligned}$$

This can also be written as $\sigma(a_k) = a_{k+1 \pmod m}$ for $k = 1, \cdots, m$, letting $a_0 = a_m$. Then, it is easy to see by induction that $\sigma^i(a_k) = a_{k+i \pmod m}$, though we won't spell out the details.

Observe that $\sigma^m = \text{id}$, since for any k we have $\sigma^m(a_k) = a_{k+m \pmod m} = a_{k \pmod m}$, and since σ will leave all other elements fixed. Also, we see that for $i = 1, \cdots, m-1$, we have $\sigma^i(a_1) = a_{1+i}$. This says that σ^i is distinct for $i = 1, \cdots, m-1$. From this we can conclude that the order of σ is m. ■

11. By Exercise 1.3.10, we know that $\sigma^i(k) = k + i \pmod m$. Then, we see that σ^i contains $(1\ 1+i\ 1+2i\ \cdots)$. This (sub-)cycle will have as length n for the smallest $n \in \mathbb{Z}^+$ satisfying $1 + ni = 1 \pmod m$, or simply $ni = 0 \pmod m$.

Now, if m and i are relatively prime, we see that the smallest $n \in \mathbb{Z}^+$ satisfying $ni = 0 \pmod m$ is $n = m$. So, this cycle is the only cycle present in σ^i, proving that it is an m-cycle.

Conversely, if σ^i is an m-cycle, it implies that $ki \neq 0 \pmod m$ for $k = 1, 2, \cdots, m-1$, and $ki = 0 \pmod m$. From here, we will conclude that i and m are relatively prime; Suppose for the sake of contradiction that $\gcd(i, m) = d > 1$, so that $i = dj$ and $m = dl$ for some $j < i$ and $l < m$. Then $il = jdl = jm$ so that $il = 0 \pmod m$. But here, since $l < m$, $il = 0 \pmod m$ contradicts that σ^i is an m-cycle, and so i and m are relatively prime. ■

15. Pick any element σ in S_n. Then the cycle decomposition of σ is written in disjoint cycles. Let us write $\sigma = c_1 c_2 \cdots c_k$ and understand it as a cycle decomposition of σ. Then, each c_i is a cycle, and they are pairwise disjoint.

Since dinjoint cycles commute, by Exercise 1.1.24, we see that $\sigma^m = c_1^m c_2^m \cdots c_k^m$ for any positive integer m. Also, by Exercise 1.3.10 we see that for any cycle c_i, c_i^m is the identity if and only if m is a multiple of the length of the cycle.

Now, for σ^m to be the identity, each of c_i^m must be the identity. (This is because the cycles are disjoint, and therefore the value of some number n under σ is only determined by at most one c_i.) Therefore, the smallest m that makes σ^m the identity must be the least common multiple of the lengths of c_i. This completes our agrument. ■

1.4 Matrix Groups

Skipped problems: 5, 6, 8∼11

1. By Exercise 1.4.7, we have $2^4 - 2^3 - 2^2 + 2 = 6$. ■

2. These 6 matrices below are nonsingular and different from one another. Since we've proved that $|GL_2(\mathbb{F}_2)| = 6$, these must be the 6 elements of the group.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

3. We can do this by simply observing that

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

while

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

which shows that the two elements do not commute. ■

4. If $\mathbb{Z}/n\mathbb{Z}$ were a field, then $(\mathbb{Z}/n\mathbb{Z} \setminus \{\overline{0}\}, \cdot)$ must be an abelian group. However, since n is not prime, there are some $p, q \in \mathbb{Z}$ such that $1 < p, q < n$ and $pq = n$. Then, for the equivalence classes $\overline{p}, \overline{q} \in \mathbb{Z}/n\mathbb{Z}$, we get $\overline{pq} = \overline{0}$. This shows that $(\mathbb{Z}/n\mathbb{Z} \setminus \{\overline{0}\}, \cdot)$ is not closed under multiplication, and thus not a group. ■

7. The total number of 2×2 matrices whose elements are from $\mathbb{F}_p$ is p^4.

Now, count the number of *singular* matrices. Let r_1 and r_2 be the two rows of the matrix.

- If $r_1 = 0$, then no matter the choice of r_2, the matrix is singular. In this case, there are p^2 singular matrices.
- Now, if $r_1 \neq 0$, then the matrix is singular if and only if $\lambda r_1 = r_2$ for some $\lambda \in \mathbb{F}_p$. Since p is prime, for each distinct λ, λr_1 are distinct. So, we have $p^2 - 1$ choices of r_1, and p choices of r_2 for each r_1, and therefore in this case there are $p^3 - p$ singular matrices.

In conclusion, there are $p^4 - p^2 - (p^3 - p) = p^4 - p^3 - p^2 + p$ nonsingular 2×2 matrices. This concludes our argument. ■

1.5 The Quaternion Group

Skipped problems: 3

1.
- 1 has order 1.
- -1 has order 2, illustrated in $-1 \to 1$.
- i has order 4, illustrated in $i \to -1 \to -i \to 1$.
- $-i$ has order 4, illustrated in $-i \to -1 \to i \to 1$,

j and k follow the exact same calculations for i, and $-j$ and $-k$ follow the exact same calculations for $-i$. Therefore they all have order 4.

2. Sure..... Here are the tables (I asked Gemini to fill in the tables)

S_3	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
1	1	(1 2)	(1 3)	(2 3)	(1 2 3)	(1 3 2)
(1 2)	(1 2)	1	(1 3 2)	(1 2 3)	(2 3)	(1 3)
(1 3)	(1 3)	(1 2 3)	1	(1 3 2)	(1 2)	(2 3)
(2 3)	(2 3)	(1 3 2)	(1 2 3)	1	(1 3)	(1 2)
(1 2 3)	(1 2 3)	(1 3)	(2 3)	(1 2)	(1 3 2)	1
(1 3 2)	(1 3 2)	(2 3)	(1 2)	(1 3)	1	(1 2 3)

D_8	1	r	r^2	r^3	s	sr	sr^2	sr^3
1	1	r	r^2	r^3	s	sr	sr^2	sr^3
r	r	r^2	r^3	1	sr^3	s	sr	sr^2
r^2	r^2	r^3	1	r	sr^2	sr^3	s	sr
r^3	r^3	1	r	r^2	sr	sr^2	sr^3	s
s	s	sr	sr^2	sr^3	1	r	r^2	r^3
sr	sr	sr^2	sr^3	s	r^3	1	r	r^2
sr^2	sr^2	sr^3	s	sr	r^2	r^3	1	r
sr^3	sr^3	s	sr	sr^2	r	r^2	r^3	1

Q_8	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	1	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

3.

1.6 Homomorphisms and Isomorphisms

Skipped problems: 4, 6, 9∼12, 17∼19, 24∼26 Unfinished problems: 23

1. a. We prove by induction that $\varphi(x^n) = \varphi(x)^n$ for all positive integers n.

If $n = 1$, it is obvious that $\varphi(x^1) = \varphi(x)^1 = \varphi(x)$. Now, for the inductive step, suppose that $\varphi(x^k) = \varphi(x)^k$ for some $k \in \mathbb{Z}^+$. Then, we see that

$$\begin{aligned}\varphi(x^{k+1}) &= \varphi(x^k x) \\ &= \varphi(x^k)\varphi(x) \quad \text{(since } \varphi \text{ is a homomorphism)} \\ &= \varphi(x)^k\varphi(x) \quad \text{(by the induction hypothesis)} \\ &= \varphi(x)^{k+1}.\end{aligned}$$

This completes our argument. ■

b. The cases where n is a positive integer is covered in part a.

First consider the case $n = 0$. This is essentially proving $\varphi(1_G) = 1_H$. This is evident once we observe that $\varphi(1_G) = \varphi(1_G \cdot 1_G) = \varphi(1_G) \cdot \varphi(1_G)$, then applying the (left-)cancellation law.

Now, consider the case $n = -1$. Observe that $1_H = \varphi(xx^{-1}) = \varphi(x)\varphi(x^{-1})$ and $1_H = \varphi(x^{-1}x) = \varphi(x^{-1})\varphi(x)$. This illustrates that $\varphi(x^{-1})$ is the inverse of $\varphi(x)$, so that $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Finally, consider the case where n is a negative integer. We wish to show that $\varphi(x^n) = \varphi(x)^n$. First, we can write $n = (-1)m$ for some positive integer m. Then we see that $\varphi(x^n) = \varphi((x^{-1})^m) = \varphi(x^{-1})^m = \varphi(x)^{-m} = \varphi(x)^n$, applying the result for when $n = -1$ or when n is positive, whenever necessary. This completes our argument. ■

From the exercise above, we can roughly say that homomorphisms maps the identity to the identity, inverses to inverses, and powers to powers.

2. Pick any $x \in G$.

First, suppose the order of x is finite and let $|x| = n$. Then, it is easy to see that $\varphi(x)^n = \varphi(x^n) = \varphi(1_G) = 1_H$.

Now, consider $\varphi(x)^k = \varphi(x^k)$ for some $k \in \{1, \cdots, n-1\}$, and we wish to show that $\varphi(x^k) \neq 1_H$. Indeed, if $\varphi(x^k) = 1_H$, then since φ is bijective we have $x^k = \varphi^{-1}(1_H) = 1_G$, which contradicts that x has order n. Therefore, we can conclude that $|\varphi(x)| = n$.

Suppose now that x has infinite order in G. Then we can use the same logic from the preceding paragraph, but where k is chosen from any of $\mathbb{Z}^+$, to say that $\varphi(x)$ also has infinite order.

From our argument, it follows that if G and H are isomorphic, then for each $n \in \mathbb{Z}^+$, the isomorphism φ restricts to a bijection from the set of elements of order n in G to the set of elements of order n in H. Hence G and H have the same number of elements of order n.

This is not true in general when φ is merely a homomorphism. Namely, the mapping $\varphi : G \to H$ that sends every element in G to 1_H is a homomorphism, but G may have elements whose order is not 1. ■

3. Suppose first that $\varphi : G \to H$ is merely a homomorphism. Then, we claim that if G is abelian, surjectivity of φ ensures that H is abelian. Assume φ is surjective. Then, for any two $h, k \in H$, we can find $x, y \in G$ such that $h = \varphi(x)$ and $k = \varphi(y)$. Then, we see that

$$\begin{aligned} hk &= \varphi(x)\varphi(y) \\ &= \varphi(xy) && \text{(since } \varphi \text{ is a homomorphism)} \\ &= \varphi(yx) && \text{(since } G \text{ is abelian)} \\ &= \varphi(y)\varphi(x) && \text{(since } \varphi \text{ is a homomorphism)} \\ &= kh. \end{aligned}$$

Since the choice of $h, k \in H$ were arbitrary, we can conclude that H is abelian.

We additionally note that if φ is not surjective then H may be non-abelian. A simple example is $\varphi : G \to H$ sending every element in G to 1_H.

Now, suppose $\varphi : G \to H$ is a group isomorphism. Then, if G is abelian then H is also abelian by surjectivity of φ. Conversely, if H is abelian then G is also abelian by surjectivity of φ^{-1}. This completes our argument. ■

5. This is evident by the fact that $|\mathbb{R}| \neq |\mathbb{Q}|$, so that there does not exist a bijection between the two. ■

7. D_8 has 5 elements of order 2, while Q_8 only has 1. Then by Exercise 1.6.2, they are not isomorphic. ■ (Compare the calculations in Exercise 1.2.1(b) and Exercise 1.5.1)

8. If $n \neq m$, then $n! \neq m!$ so $|S_n| \neq |S_m|$. Therefore we cannot construct a bijection between the two groups. ■

13. It is obvious that $\varphi(G)$ is nonempty. Now, pick any $h, k \in \varphi(G)$. Then, there exist some $x, y \in G$ such that $\varphi(x) = h$ and $\varphi(y) = k$. Then, we have

$$hk = \varphi(x)\varphi(y) = \varphi(xy) \in \varphi(G),$$

and

$$h^{-1} = \varphi(x)^{-1} = \varphi(x^{-1}) \in \varphi(G).$$

So, $\varphi(G)$ is closed under multiplication and inverses. This concludes that $\varphi(G)$ is a subgroup of H. ■

14. We first see that $\ker\varphi$ is nonempty, since in particular $\varphi(1_G) = 1_H$. Now, pick any $g, h \in \ker\varphi$. Then, we see that $\varphi(gh) = \varphi(g)\varphi(h) = 1_H \cdot 1_H = 1_H$ and $\varphi(g^{-1}) = \varphi(g)^{-1} = 1_H^{-1} = 1_H$. This shows that $gh, g^{-1} \in \ker\varphi$, concluding that $\ker\varphi$ is a subgroup of G.

Now, we wish to show that $\ker\varphi = \{1_G\}$ if and only if φ is injective. The if direction is evident, so for the other direction suppose that $\ker\varphi = \{1_G\}$. Suppose also that $\varphi(x) = \varphi(y)$. We want to show that $x = y$. Here, we see that

$$1_H = \varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}).$$

But since $\ker\varphi$ has the only element 1_G, $\varphi(xy^{-1}) = 1_H$ implies that $xy^{-1} = 1_G$, or that $x = y$, as required. This completes our argument. ■

15. This is immediate from Exercise 1.6.16.

16. To see that π_1 is a homomorphism, observe that for any $(a_1, b_1), (a_2, b_2) \in G$, we have

$$\begin{aligned}\pi_1((a_1, b_1)(a_2, b_2)) &= \pi_1(a_1a_2, b_1b_2) \\ &= a_1a_2 \\ &= \pi_1(a_1, b_1)\pi_1(a_2, b_2).\end{aligned}$$

This shows that π_1 is a homomorphism.

An element (a, b) in G is in the kernel of π_1 if and only if $\pi_1((a, b)) = 1_A$, which only happens if and only if $a = 1_A$. Therefore, the kernel can be written as $\{1_A\} \times B$.

Things about π_2 can be proven in the exact same steps. ■

20. Pick any $f, g, h \in \mathrm{Aut}(G)$. We first show that $\mathrm{Aut}(G)$ is closed under function composition, that is, fg is an isomorphism from G onto G. To see that fg is a homomorphism, observe that for any $x, y \in G$ we have $fg(xy) = f(g(x)g(y)) = fg(x)fg(y)$, since f and g are both homomorphisms. Also, since f and g are both bijective, fg is also bijective. This shows that $fg \in \mathrm{Aut}(G)$.

Now, it suffices to check the three group axioms.

- Associativity: Since function composition is associative, we see that $(fg)h = f(gh)$.
- Identity: The identity function id_G is the identity. It is obvious that $f\,\mathrm{id}_G = \mathrm{id}_G\, f = f$, so we are left to first verify that $\mathrm{id}_G \in \mathrm{Aut}(G)$; For any $x, y \in G$, we see that $\mathrm{id}_G(xy) = xy = \mathrm{id}_G(x)\,\mathrm{id}_G(y)$.
- Inverse: For f, f^{-1} satisfies $ff^{-1} = f^{-1}f = \mathrm{id}_G$. It is left to check that $f^{-1} \in \mathrm{Aut}(G)$. For any $x, y \in G$, we have

  $$f(f^{-1}(x)f^{-1}(y)) = f(f^{-1}(x))f(f^{-1}(y)) = xy.$$

  Applying f^{-1} on both sides gives $f^{-1}(x)f^{-1}(y) = f^{-1}(xy)$, hence f^{-1} is an isomorphism from G onto G.

This completes our argument. ■

21. View $\mathbb{Q}$ as an additive group. Fix any $k \in \mathbb{Q}$. Let $\sigma : \mathbb{Q} \to \mathbb{Q}$ be the mapping by $\sigma(q) = kq$. We wish to show that σ is a homomorphism, and that it is bijective.

To see that σ is a homomorphism, observe that for any $p, q \in \mathbb{Q}$,

$$\begin{aligned}\sigma(p + q) &= k(p + q) \\ &= kp + kq \qquad \text{(distributivity of } \mathbb{Q}) \\ &= \sigma(p) + \sigma(q).\end{aligned}$$

To see that σ is bijective, observe that the mapping $\tau : \mathbb{Q} \to \mathbb{Q}$ defined by $\tau(q) = \frac{1}{k} \cdot q$ is an explicit inverse (noting that $\tau(q) \in \mathbb{Q}$ is indeed true):

$$\tau\sigma(q) = \tau(kq) = \frac{1}{k}kq = q$$

and

$$\sigma\tau(q) = \sigma\left(\frac{1}{k}q\right) = k\frac{1}{k}q = q.$$

Therefore, $\tau = \sigma^{-1}$ and σ is bijective. This completes our argument. ■

22. Let A be an abelian group, and fix some $k \in \mathbb{Z}$. We wish to show that $\sigma : A \to A$ defined by $\sigma(a) = a^k$ is a homomorphism. To see this, observe that for any $a, b \in A$,

$$\begin{aligned} \sigma(ab) &= (ab)^k \\ &= a^k b^k && (\because \text{Exercise 1.1.24}) \\ &= \sigma(a)\sigma(b). \end{aligned}$$

Now, let $k = -1$, and we wish to see that in this case σ is an isomorphism. To do this, it suffices to show that σ is bijective. We claim that σ itself is the inverse of σ; see that

$$\sigma\sigma(a) = \sigma(a^{-1}) = (a^{-1})^{-1} = a.$$

This completes our argument. ■

23. Let G be finite, and σ be an automorphism of G such that $\sigma(g) \neq g$ whenever $g \neq 1$, and $\sigma^2 = \mathrm{id}_G$. Following the hint, we will first show that every element of G can be written in the form $x^{-1}\sigma(x)$.

First, observe that $x \neq y$ implies that $x^{-1}\sigma(x) \neq y^{-1}\sigma(y)$. This is because

1.7 Group Actions

Skipped problems: 2, 3, 5, 6, 8∼10, 12, 20∼23

1. We first see that for any element a in the field, $a0 = 0$. This is because $a0 = a(0+0) = a0 + a0$ by distributivity, and then $a0 = 0$ by the cancellation laws. Similarly, $0a = 0$.

Now, we prove the two group action axioms.

1. $g_1(g_2a) = (g_1g_2)a$: Pick any $g_1, g_2 \in F^\times$.
   - If $a = 0$, then viewing every operation as multiplication in F, we see that $g_1(g_2a) = g_10 = 0$ and $(g_1g_2)a = (g_1g_2)0 = 0$, so that they are equal.
   - Otherwise, we can view every operation as multiplication in $F^\times$. Then, the property directly follows from the associativity of $F^\times$.
2. $1a = a$: If $a = 0$, then $1a = 0$ in F. Else, $1a = a$ in F.

This completes our argument. ■

4. To see that H is a subgroup of G, it suffices to show that for any $h, k \in H$, we have $hk \in H$ and $h^{-1} \in H$.

a. Let $H = \{g \in G \mid gb = b \text{ for all } b \in A\}$. Pick any $h, k \in H$. Then, for any $b \in A$, we have

$$\begin{aligned}(hk)b &= h(kb) && \text{(group action axiom)}\\ &= hb && (k \in H)\\ &= b, && (h \in H)\end{aligned}$$

which shows that $hk \in H$. Also,

$$\begin{aligned}h^{-1}b &= h^{-1}(hb) && (h \in H)\\ &= (h^{-1}h)b && \text{(group action axiom)}\\ &= b, && \text{(group action axiom)}\end{aligned}$$

which shows that $h^{-1} \in H$. ■

b. Now fix some $a \in A$, and let $H' = \{g \in G \mid ga = a\}$. This is exactly the set

$$\{g \in G \mid gb = b \text{ for all } b \in \{a\}\},$$

that is, the subgroup from part (a) with A replaced by the singleton $\{a\}$. Therefore, following the same logic as in part (a), H' is a subgroup of G. ■

7. We have the group $\mathbb{R}$ acting on the set $\mathbb{R}^n$ where the action is

$$\alpha(r_1, \cdots, r_n) = (\alpha r_1, \cdots, \alpha r_n)$$

for $\alpha \in \mathbb{R}$ and $(r_1, \cdots, r_n) \in \mathbb{R}^n$. To show that this action is faithful, we will show that the kernel is trivial. Namely, assume

$$\alpha(r_1, \cdots, r_n) = (r_1, \cdots, r_n)$$

for any $(r_1, \cdots, r_n) \in \mathbb{R}^n$, and we will show that this implies $\alpha = 1$.

From the previous assumption we have $\alpha r_i = r_i$ for $i = 1, \cdots, n$ where each r_i is an arbitrary element in $\mathbb{R}$. Since $\mathbb{R}$ is a field, then it must be that $\alpha = 1$, as required. ■

11. Recall from Section 1.2 that the vertices of the square are labeled $1, 2, 3, 4$ in clockwise order. Let r be the clockwise rotation by $2\pi/4$, and let s be the reflection across the line through vertex 1 and the center of the square.

As permutations of $\{1, 2, 3, 4\}$, we have $r = (1\ 2\ 3\ 4)$ and $s = (2\ 4)$.

Therefore, the eight elements of D_8 are:

$$\begin{aligned} 1 &= \mathrm{id}, \\ r &= (1\ 2\ 3\ 4), \\ r^2 &= (1\ 3)(2\ 4), \\ r^3 &= (1\ 4\ 3\ 2), \\ s &= (2\ 4), \\ sr &= (1\ 4)(2\ 3), \\ sr^2 &= (1\ 3), \\ sr^3 &= (1\ 2)(3\ 4). \end{aligned}$$

13. For a group G acting on the set G, we must determine what the set

$$\{g \in G \mid ga = a \text{ for all } a \in G\}$$

looks like. But in a group, $ga = a$ implies $g = 1$ by right cancallation. Therefore, the kernel must be the trivial subgroup $\{1\}$. ■

(This shows that the left regular action is faithful.)

14. Let G a group and $A = G$. If the maps are defined by $g \cdot a = ag$, then it fails to satisfy the compatibility axiom. To see this, pick any $g, h \in G$ and $a \in A$. We have

$$(gh) \cdot a = a(gh),$$

but

$$g \cdot (h \cdot a) = g \cdot (ah) = (ah)g.$$

For compatibility to be satisfied, the two must equal, so that $a(gh) = (ah)g$. But then left cancellation implies $gh = hg$, which is exactly the abelian condition. Therefore, this map satisfy the compatibility axiom if and only if G is abelian, or in other words if G is non-abelian the map does *not* satisfy the axiom. ■

15. Let G a group, $A = G$, and define the maps by $g \cdot a = ag^{-1}$. Then, fix arbitrary $g, h \in G$ and $a \in A$. We then see that

$$(gh) \cdot a = a(gh)^{-1} = ah^{-1}g^{-1}$$

and

$$g \cdot (h \cdot a) = g \cdot (ah^{-1}) = (ah^{-1})g^{-1},$$

so that $(gh) \cdot a = g \cdot (h \cdot a)$. Also, $1 \cdot a = a1^{-1} = a$. This completes our argument. ■

16. Let G a group, $A = G$, and define the maps by $g \cdot a = ag^{-1}$. Then, fix arbitrary $g, h \in G$ and $a \in A$. Then, we see that

$$(gh) \cdot a = (gh)a(gh)^{-1} = ghah^{-1}g^{-1}$$

and

$$g \cdot (h \cdot a) = g \cdot (hah^{-1}) = g(hah^{-1})g^{-1}.$$

From this we observe that $(gh) \cdot a = g \cdot (h \cdot a)$. Also, $1 \cdot a = 1a1^{-1} = a$. This completes our argument. ■

17. Let G a group, and fix an element $g \in G$. Define a map $\varphi : G \to G$ by

$$\varphi(x) = gxg^{-1}.$$

We will show that φ is an isomorphism.

First, check the homomorphism property. For any $x, y \in G$, we see that

$$\varphi(xy) = g(xy)g^{-1},$$

while

$$\varphi(x)\varphi(y) = (gxg^{-1})(gyg^{-1}) = gxyg^{-1},$$

so that $\varphi(xy) = \varphi(x)\varphi(y)$.

Now, to see that φ is bijective, we will show that the map $\psi : G \to G$ defined by

$$\psi(x) = g^{-1}xg$$

is a two-sided inverse of φ. Indeed, we can easily see that

$$(\psi \circ \varphi)(x) = \psi(gxg^{-1}) = g^{-1}(gxg^{-1})g = x$$

and

$$(\varphi \circ \psi)(x) = \varphi(g^{-1}xg) = g(g^{-1}xg)g^{-1} = x.$$

This completes our argument that φ is an isomorphism from G onto itself. ■

18. For the relation defined in the exercise, we must show reflexivity, symmetry, and transitivity.

- Reflexivity: Fix any $x \in A$. Since $1 \in H$, we have $x = 1x$ so that $x \sim x$.
- Symmetry: Suppose $x \sim y$ for some $x, y \in A$. Then we have $x = hy$ for some $h \in H$. This implies $y = h^{-1}x$, and since $h^{-1} \in H$ this shows that $y \sim x$.
- Transitivity: Suppose $x \sim y$ and $y \sim z$ for some $x, y, z \in A$. Then we have $x = hy$ and $y = kz$ for some $h, k \in H$. From this we can deduce $x = h(kz) = (hk)z$, and since $hk \in H$ we conclude that $x \sim z$.

This shows that the relation is indeed an equivalnce relation. ■

19. Let G be a group and H a subgroup of G. Fix an element $x \in G$. Then, the orbit $\mathcal{O}$ of x can be written by

$$\mathcal{O} = \{hx \mid h \in H\}.$$

Now, define a map $\varphi : H \to \mathcal{O}$ by $\varphi(h) = hx$. We wish to show that φ is bijective.

First, to see surjectivity, pick any $g \in \mathcal{O}$. Then, by definition of the orbit, we have $g = hx$ for some $h \in H$. Therefore, for this h we have $\varphi(h) = g$, proving surjectivity.

Now, to see injectivity, suppose $\varphi(h) = \varphi(k)$ for some $h, k \in H$. This implies that $hx = kx$, and right cancellation in G yields $h = k$. This proves injectivity. Therefore, φ is bijective, so the orbit of x has cardinality $|H|$. Since $x \in G$ was arbitrary, every orbit has cardinality $|H|$.

By Exercise 1.7.18, the orbits under the action of H partition the set G. Thus G is a disjoint union of orbits, each of cardinality $|H|$. Therefore, if $|G|$ is finite then it must be that $|H|$ divides $|G|$. ■